



NOTIFICATION AND COMMUNICATION SUBSYSTEM

**Presented to the Faculty of Information Technology
School of Engineering, Computer Studies and Architecture**

In Partial Fulfillment

**Of the Requirements for the Degree of
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

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May 2026

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Abstract

Write your abstract here, in one whole page

Chapter 1

Introduction

1.1 Background

Effective communication in educational settings is important for maintaining discipline, providing timely updates, and improving student monitoring. In Computer Laboratory A,

attendance monitoring is a key activity [1], but traditional methods often rely on manual recording and delayed communication. This can result in errors, late reporting, and lack of immediate information for guardians or parents regarding student attendance [2].

To address these issues, this study proposes the design and development of a Notification and Communication Subsystem specifically for Computer Laboratory A. The system sends real-time notifications to students' guardians or parents regarding attendance status, such as present or absent [3].

The subsystem includes several components that ensure notifications are accurate, timely, and meaningful. One of its key features is the ability to recognize and differentiate student Barcode scans, which helps prevent errors in recording attendance. By focusing specifically on notification and communication, the system improves how attendance information is delivered to parents or guardians in real time [4]. This allows them to stay informed about their child's presence in class. In addition, the subsystem supports better monitoring and management of students in Computer Laboratory A, making the overall attendance process more reliable and efficient.

1.2 Statement of the Problem

This study focuses on improving notification and communication processes in Computer Laboratory A, particularly in informing guardians or parents about student attendance. Existing manual methods may cause delays, errors, and lack of timely updates.

The study is guided by the following research concerns:

- How a Notification and Communication Subsystem can improve the delivery of attendance information to guardians or parents in Computer Laboratory A.
- The features required to ensure notifications are sent only when appropriate and not triggered by teacher barcode scans.
- The ways in which the system can ensure notifications are timely, accurate, and relevant.
- The impact of the subsystem on communication efficiency and parental awareness in Computer Laboratory A.
- The study aims to develop a system that provides reliable and real-time attendance notifications to improve communication.

1.3 Objective of the Study

General Objectives

To design and develop a Notification and Communication Subsystem that improves the delivery of attendance notifications to guardians or parents in Computer Laboratory A.

Specific Objectives

- To implement a system that sends real-time attendance notifications to guardians or parents.
- To ensure notifications are triggered only by student barcode scans and not by teachers.
- To improve communication efficiency by automating attendance notifications and reducing manual processes.
- To evaluate the effectiveness of the subsystem in improving communication and parental

awareness in Computer Laboratory A.

Significance of the Study

The development of the Notification and Communication Subsystem is significant because it enhances communication and monitoring efficiency within computer laboratory A by ensuring that guardians or parents receive timely and accurate notifications regarding student attendance.

Students

The system promotes accountability and awareness of their attendance and class schedules, supporting their academic responsibilities.

For **guardians or parents**, the subsystem provides real-time updates about their child's attendance, helping them stay informed and engaged in their child's academic progress.

For **Teacher** workload while improving the efficiency of conveying information to guardians and students.

Overall, the subsystem fosters timely and reliable communication, strengthens parental engagement, and supports effective academic monitoring, ultimately contributing to better student discipline and performance.

1.4 Scope and Limitations

Scope

This study focuses on the design and development of a Notification and Communication Subsystem implemented only in Computer Laboratory A. The system is designed to send attendance notifications to parents or guardians based on student presence. It uses Barcode scanning to record attendance, where only student scans trigger notifications while teacher scans are filtered. The subsystem includes real-time data processing and uses Application Programming Interfaces (APIs) to communicate with external email or messaging services. Attendance data is stored in a secure database to ensure proper record management.

Limitations

The system presents several limitations that may affect its functionality and scope. It does not incorporate scheduling management features and is restricted exclusively to Computer Laboratory A. Its operation depends on a functioning SIM network connection and SMS gateway services to transmit notifications. Consequently, performance may be hindered if the mobile signal is weak, the SIM card lacks sufficient load or balance, or the SMS service experiences temporary interruptions. Notification delays may also occur when multiple users attempt to scan simultaneously.

Another limitation involves the potential risk of barcode misuse, such as sharing or unauthorized scanning, which can compromise the accuracy of attendance data. Furthermore, the system has been designed specifically for Computer Laboratory A and may require modifications if implemented in other areas of the institution.

And the system is offline-based, as it utilizes SIM card technology for SMS communication. Internet connectivity is not required for guardians or parents to receive attendance notifications, ensuring accessibility even in environments with limited online infrastructure.

Definition of terms

Barcode Code - a machine-readable code consisting of an array of black and white squares, typically used for storing URLs or other information for reading by the camera on a smartphone.

- **Barcode Scanner**-a device which is used to scan the QR codes
- **Smart Phone** - a mobile phone that performs many of the functions of a computer, typically having a touchscreen interface, Internet access, and an operating system capable of running downloaded applications.
- **SMS** - Short Message Service and is the most widely used type of text messaging.
- **Automated**- operated by largely or small automatic equipment.
- **Encrypted**- convert (information or data) into a cipher or code.

Chapter 2

Review of Related Literature

2.1 Related Works

Foreign studies

Modern notification systems emphasize real-time alerts as a deterrent to absenteeism [1]. The transition from manual to electronic logging laid the foundation for digital attendance systems [2]. Automated management and alert systems have been highlighted as essential for maintaining

discipline in academic institutions [3].

The introduction of iBeacon technology enabled validation of student presence within classroom boundaries, ensuring geographically accurate notifications [4]. The integration of IoT and RFID technologies created smart environments where attendance is captured seamlessly without human intervention [5].

RFID-based systems remain the gold standard for efficient record management due to their speed and reliability [6]. Intelligent notification systems emphasize context-aware computing to ensure relevance and minimize disruption [7].

The robustness of automated notification systems has also been demonstrated in other sectors, such as wireless sensor networks for water quality monitoring [8]. GPS-based monitoring has been applied for infrastructure, showing cross-sector adaptability [9]. More recently, communication strategies in learning reinforced the role of notifications in bridging institutional and parental engagement [10].

Local Studies

Web-based attendance systems with SMS notifications have proven effective in public high schools, providing centralized platforms capable of handling large student datasets [11]. Event management systems with SMS capabilities have been successfully implemented in Mindanao, proving adaptability in regional contexts [12].

RFID-based attendance systems incorporating data analytics allow administrators to identify

attendance trends and provide proactive support for at-risk students [13]. QR codes integrated with SMS notifications have been shown to be scalable and cost-effective for basic education units [14]. Android-based classroom monitoring systems using QR technology emphasize the effectiveness of mobile-based solutions in the Philippine setting, where smartphone use is widespread among educators [15]. emphasizing that mobile-based solutions are particularly effective in the Philippine setting where smartphone use is ubiquitous among educators. Together, these local studies suggest a strong movement toward affordable, mobile-integrated, and data-driven attendance solutions.

Theoretical Frameworks

The Notification and Communication Subsystem is grounded in several theoretical foundations. First, the Communication Model explains how information flows from sender to receiver through a channel, with potential barriers that may distort the message. Applied to attendance systems, the student barcode scan acts as the sender, attendance data serves as the message, the SMS gateway functions as the channel, and the parent or guardian is the receiver. This framework underscores the importance of minimizing interference such as weak signals or misuse of barcodes [5].

The Notifications Framework emphasizes event-driven and context-aware communication. This ensures that alerts are triggered only when appropriate, thereby preventing false notifications such as those caused by teacher barcode scans. By applying this principle, the subsystem maintains relevance and avoids information overload [6].

The Technology Acceptance Model (TAM) provides a lens for understanding how users

adopt and interact with new systems. TAM posits that perceived usefulness and perceived ease of use are the primary factors influencing user acceptance of technology. In the subsystem, teachers and parents are more likely to adopt the system if they believe it improves attendance monitoring and is simple to operate [7].

Innovation of Systems (IOS) framework highlights how new technologies are integrated into existing institutional processes. IOS emphasizes adaptability, scalability, and alignment with organizational goals. For the subsystem, this means ensuring that the notification process fits seamlessly into school operations while remaining flexible enough to accommodate future upgrades [8].

Context-Aware Computing and Human-Computer Interaction (HCI) theories emphasize timeliness and relevance of notifications by considering the context in which communication occurs. This ensures that parents receive meaningful updates about their child's attendance while irrelevant events are filtered out [9].

The subsystem uses Application Programming Interfaces (APIs) to connect the system to external messaging services. These APIs allow the system to automatically send notifications once attendance is recorded. To ensure secure communication, the system uses encrypted connections (such as HTTPS), protecting the data while it is being transmitted.

Together, these frameworks provide a strong theoretical basis for the design and implementation of the Notification and Communication Subsystem. They explain how information can be transmitted effectively, how technology adoption is facilitated, and how innovation aligns

with institutional needs.

Privacy

Privacy and data protection are also considered in this study. The subsystem follows data privacy principles by limiting the collection of personal data to only what is necessary, such as student name and guardian contact details. Access to data is restricted, and sensitive information is protected to prevent unauthorized use.

Security

Security is crucial for the Notification and Communication Subsystem in Computer Laboratory A, which deals with sensitive data like student attendance and guardian contact details. It employs encryption protocols like TLS/SSL for secure data transmission, preventing unauthorized access. User authentication restricts access to authorized personnel, while validation mechanisms ensure notifications are only triggered by valid student Barcode scans, avoiding false alerts. The system's limitation to Computer Laboratory A further enhances security by controlling access and mitigating risks. Overall, these measures ensure data privacy, reliability, and secure communication with guardians.

Prevention

Prevention mechanisms are included to ensure accurate and relevant notifications. The system filters Barcode scans to distinguish between students and teachers, preventing incorrect alerts. It also avoids duplicate notifications by checking if attendance has already been recorded. These measures help reduce errors and ensure that only valid and necessary information is communicated.

By applying these theories, along with secure API integration, data privacy measures, and

prevention mechanisms, the subsystem improves real-time communication, enhances parental awareness, and supports effective attendance monitoring specifically within Computer Laboratory A.

Conceptual Frameworks

The conceptual framework of the subsystem illustrates how student attendance data is collected, processed, and communicated to guardians or parents to improve academic monitoring and communication efficiency.

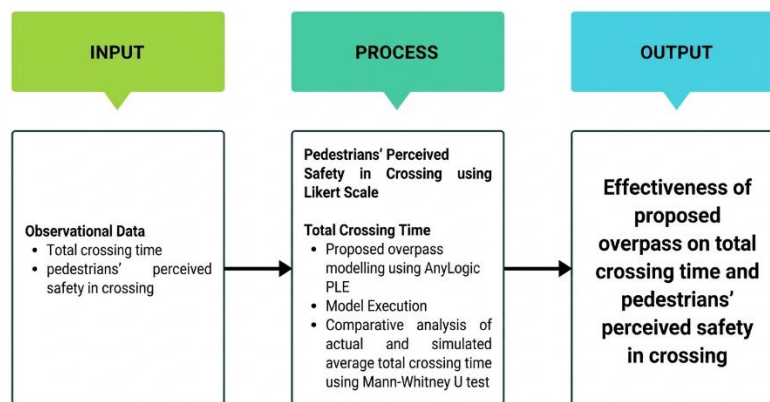


Figure 1

2.1.1 Input

The inputs of the system include student barcode scans for attendance and the contact details of guardians or parents for message delivery. These inputs are processed by the system to ensure that notifications are sent only when appropriate, preventing unnecessary alerts such as those triggered by teacher scans. Through this process, the system ensures that guardians receive timely, relevant, and accurate information regarding student attendance and schedules.

2.1.2 Process

The system processes the collected inputs through several steps. First, it scans and verifies the student barcode to confirm identity. Then, it records attendance in the database. After validation, the system checks if the scan belongs to a student and not a teacher. Once confirmed, the notification trigger module activates and prepares the message using the stored parent or guardian contact details. The system ensures that only valid and necessary notifications are generated to avoid errors or duplicate alerts.

2.2.3 Output

The output of the system is the successful delivery of attendance notifications to the registered guardians or parents. These notifications are sent through SMS, informing them whether the student is present or absent. The system also generates stored attendance records that can be viewed by authorized users. This output ensures that parents receive accurate, timely updates, improving communication, and allowing better monitoring of student attendance.

Chapter 3

Research Methodology

3.1 Software Life Cycle Model - Iterative Process Model

3.1 Software Development Life Cycle (SDLC)

For this study, the **Iterative Process Model** is used as the Software Development Life Cycle (SDLC). This model is appropriate because it allows the system to be built step by step.

Instead of completing everything at once, small working parts of the system are created, tested, feedback is gathered, and improvements are made in the next cycle. Since the Notification and Communication Subsystem has multiple components—such as barcode scanning, sending notifications, and managing users—the iterative model enables early issue resolution and gradual feature integration.

The phases of the model are applied as follows:

- **Planning:** The main goal of the system is defined, which is to send real-time attendance notifications to parents. Basic features are identified, including student barcode login and automated text alerts.
- **Analysis:** The current attendance recording process in Computer Laboratory A is examined. Teachers are consulted, and problems such as delays or errors in the manual process are identified.
- **Design:** A simple plan for the system is prepared. This includes drawing the database structure, designing the user interface, and determining how data flows from barcode scans to parent notifications.
- **Development:** Code is written for one small feature at a time. For example, QR code scanning for students is built first, followed by the notification-sending feature.
- **Testing:** Each feature is tested to ensure proper functionality. Notifications are checked to confirm they are sent only for student scans and not for teacher scans.
- **Deployment:** Once the system is stable and tested, it is installed in Computer Laboratory A for actual use.
- **Maintenance:** The system is continuously monitored, bugs are fixed, and improvements

are made based on teacher and parent feedback.

3.2 Research Design

This study adopts a quantitative research design to evaluate the effectiveness of the Notification and Communication Subsystem in Computer Laboratory A. The design focuses on collecting numerical data that can be measured, analyzed, and interpreted statistically. By using this approach, the study ensures objectivity and reliability in assessing system performance.

The research applies a developmental method since the primary goal is to design and build a new subsystem that addresses existing communication gaps in attendance reporting. The system is developed iteratively, and each cycle is tested to generate measurable outcomes such as notification accuracy, timeliness, and error rates.

To gather data, the study employs a survey method. Structured questionnaires are distributed to parents or guardian to collect quantifiable feedback on system usability, efficiency, and reliability. The responses are analyzed statistically to determine the impact of the subsystem on communication efficiency and parental awareness.

Target Respondents

- Who: Parents whose children are enrolled in schools using the attendance system, or parents directly affected by its notifications (e.g., receiving SMS).
- Focus: Understanding their user experience, perceived safety, and communication satisfaction.

2. Setting & Platform

- Digital Environment: The study will be conducted online.

- **Tool:** Google Forms will be used for distribution, which is excellent for automatic data organization and accessibility on mobile devices.

3. Data Gathering Instruments

Survey Questionnaires (Quantitative): Used to measure “how much” or “how often” responses through structured questions. The survey will be conducted online using Google Forms. It includes rating-scale questions such as “How satisfied are you on a scale of 1–5?” to gather measurable data on user experiences, system performance, and notification reliability.

3.3 Data Gathering Procedure

We will follow these steps to collect data for our study through digital platforms:

- **Prepare the Tools:** We will design a digital survey questionnaire and an online interview guide specifically tailored for parents.
- **Ask Permission:** We will request formal permission from the school administration to contact parents and to observe the current attendance workflow in Computer Laboratory A.
- **Conduct Observations:** We will observe how teachers currently record attendance in the lab to understand the "pain points" that eventually affect the notifications parents receive.
- **Distribute Online Surveys:** We will send the survey link (via Google Forms) to the target parents. The survey will focus on their expectations for the new system and their notification preferences.
- **Collect and Organize Data:** Data from Google Forms will be automatically organized into spreadsheets. Interview notes will be transcribed and categorized.
- **Analyze Data:** We will study the responses to identify common requirements. These results will directly inform the design of the Barcode notification system.

3.4 Research Respondents

The respondents for this study are the parents of students who utilize Computer Laboratory A.

- **Parents (10 Respondents):** They are the primary stakeholders for the notification feature. We will gather data on their preferred communication channels SMS and the value they place on real-time attendance updates.

Sampling Method: Sampling Method: We will use purposive sampling. We will specifically select parents whose children have regular classes in Computer Laboratory A to ensure that the feedback is relevant to the specific environment where the system will be implemented. Data will be collected through an online survey method only. No interviews will be conducted in this study.

3.5 Research Environment

The study focuses on the attendance issues within Computer Laboratory A, although the parents will provide their input online.

- **Context:** The attendance problem being solved exists in this specific lab.
- **Accessibility:** Since the survey is online, parents can participate from their homes or workplaces at their convenience.
- **Infrastructure:** The lab's existing internet and computer setup will be used to host and manage the digital survey responses.

3.6 Research Instrument

We will use one main instrument to gather data:

- **Online Survey Questionnaire:** This is a digital form created via Google Forms consisting of:
- **Rating Scales (1 to 5):** To measure how much parents value real-time notifications.
- **Short Response:** To allow parents to list any specific concerns they have about student tracking.

An online survey is the most appropriate tool because it allows for efficient, contactless data collection.

It automatically summarizes the numbers (quantitative data), making it easy to see exactly what the

majority of parents expect from the new system.

3.7 Software Requirements

To build and run the Notification and Communication Subsystem, we need the following software:

System Requirements (for development and hosting):

Table 1

Category	Requirement	Details
<u>Operating System</u>	Windows 10 or higher / Linux	Required for development and hosting environment
<u>Web Server</u>	XAMPP (Apache)	Runs the system locally
<u>Database</u>	MySQL	Stores student, attendance, and parent data
<u>Backend Language</u>	PHP	Processes system logic (e.g., checking absences)
<u>Frontend Languages</u>	HTML, CSS, JavaScript	Provides user interface and interaction
<u>Browser</u>	Google Chrome / modern browser	Used to access the system

User Requirements:

Table 2

Category	Requirement	Details
<u>Student Attendance Logging</u>	Barcode scanning	Students log attendance by scanning their barcode
<u>Teacher Scan Filtering</u>	No notification for teacher scans	Prevents false alerts
<u>Automated Notifications</u>	SMS text messages	Parents receive real-time notifications when their child enters Computer Laboratory A
<u>Notification Logs</u>	Stored in database	Keeps a record of all notifications sent
<u>Attendance Records</u>	Teacher access	Teachers can view attendance records by date and subject

3.8 Hardware Requirements

To develop, test, and run the system, we need the following physical devices. These are the minimum specifications for smooth performance.

Table 3

Device	Minimum Specification	Purpose
Laptop/Desktop (Developer)	Intel Core i5 processor, 4GB RAM, 256GB storage	For writing code, designing the database, and testing the system.
Desktop (School Lab)	Intel Core i5 or higher, 2GB RAM, 100GB storage	For teachers to access the dashboard and for students to scan Barcodes.
Smartphone (Student)	Android or iOS with camera	For students to scan the barcode displayed on the screen.
Smartphone (Parent)	Any phone that can receive SMS	For parents to receive notification messages.
Router/Wi-Fi	Standard 100 Mbps router	To connect all devices to the local network and internet.
Printer	Standard inkjet/laser printer	To print Barcodes or backup attendance reports.

3.9 Data Flow Diagram

Level 0 DFD (Context Diagram)

This diagram shows the whole system as one process. It shows who gives data to the system and who receives data from the system.

- External Entities: Student, Teacher, Parent
- Main Process: Notification and Communication Subsystem

Data Flows:

- Student → System: barcode Code Scan Data
- Teacher → System: Student List, Schedule Info
- System → Parent: Attendance Notification
- System → Teacher: Attendance Reports

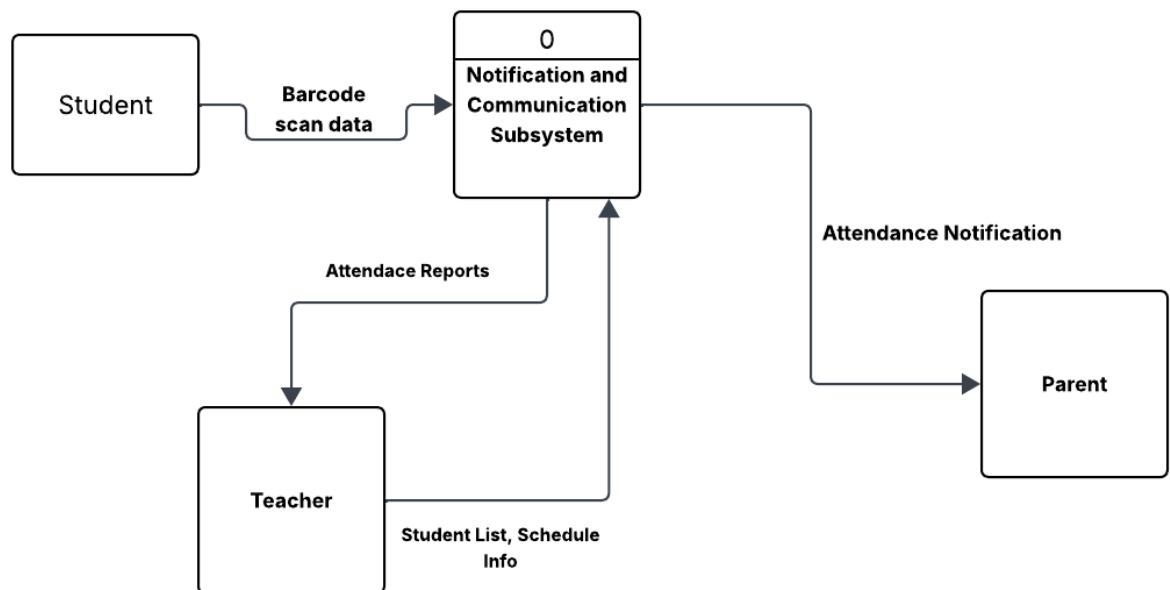


Figure 2

Level 1 DFD

This diagram breaks the main process into smaller sub-processes. It shows more detail.

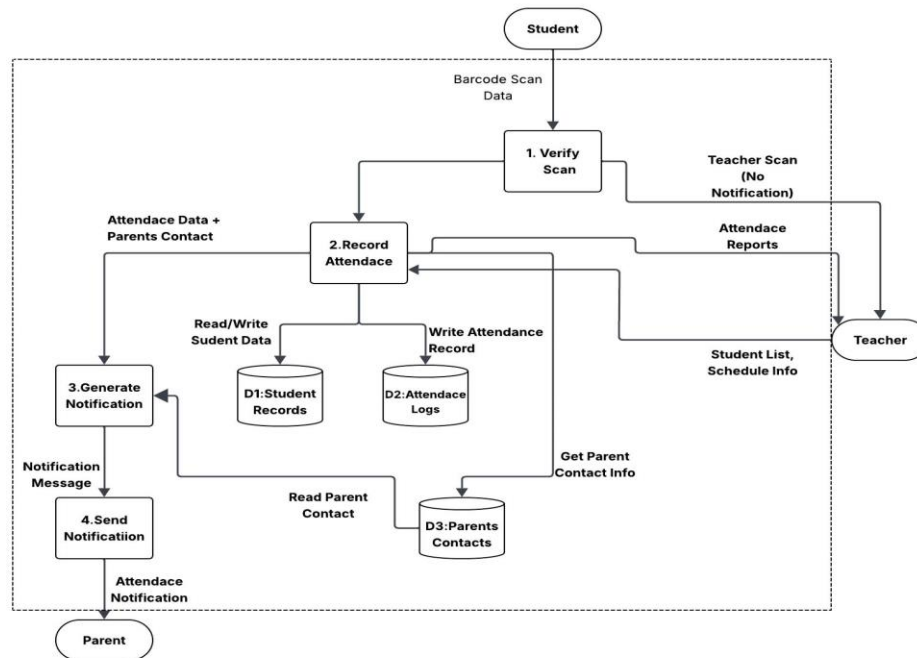


Figure 3

Sub-processes:

1. **Verify Scan:** Checks if the scan is from a student or a teacher.
2. **Record Attendance:** Saves the attendance data (time, date, student name) to the database.
3. **Generate Notification:** Creates the message content (e.g., "Your child John arrived at 8:00 AM").
4. **Send Notification:** Delivers the message to the parent's phone.

Data Stores:

- D1: Student Records
- D2: Attendance Logs

- D3: Parent Contacts

Data Flows: Show how data moves from the "Verify Scan" process to "Record Attendance," then to "Generate Notification," and finally to "Send Notification," with arrows connecting to the data stores.

3.10 Use Case Diagram

The Use Case Diagram shows the main functions (use cases) of our system and the people (actors) who will use them.

Actors:

- Student: The person who scans the Barcode.
- Teacher: The person who manages the class and views reports.
- Parent: The person who receives notifications.

Use Cases (Functions):

- Login (for Teacher)
- Scan barcode (for Student)
- Mark Attendance (System does this automatically)
- Send Notification (System does this automatically)
- Receive Notification (for Parent)
- View Attendance Report (for Teacher)

- Manage Student List (for Teacher)

Relationships:

- The students is connected to "Scan Barcode".
- The Teacher is connected to "Login", "View Attendance Report", and "Manage Student List".
- The Parent is connected to "Receive Notification".
- The "Send Notification" use case is connected to "Mark Attendance" (it extends it).

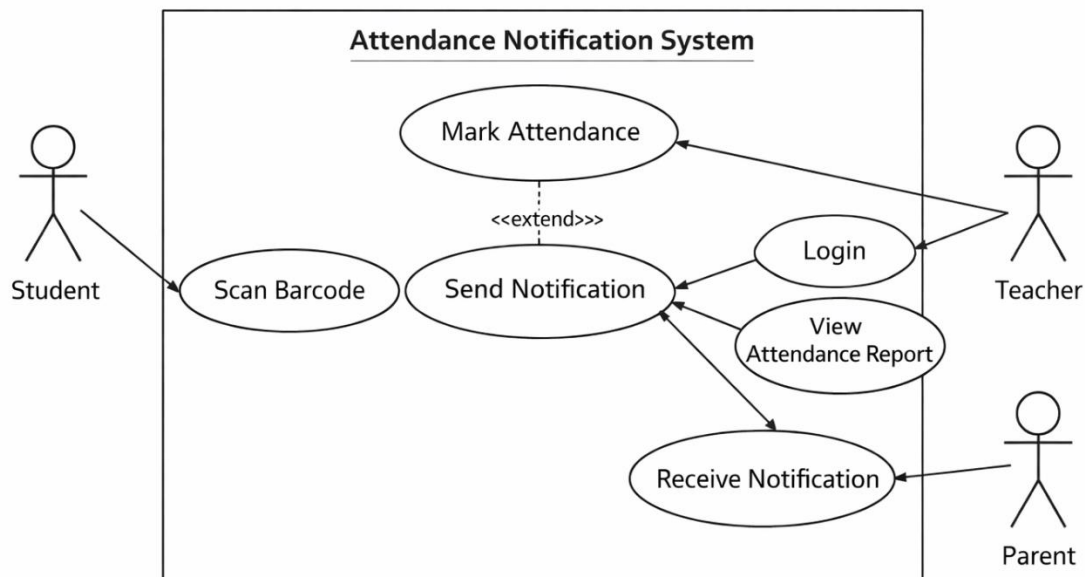


Figure 4

3.11 Data Dictionary

This section lists the main data elements (tables and fields) in our database.

Table: student

Field Name	Data Type	Size	Description	Purpose
student_id	INT	11	Unique number	Primary key to identify each student
first_name	VARCHAR	50	Student's first name	Stores the student's given name
last_name	VARCHAR	50	Student's last name	Stores the student's surname
qr_code	VARCHAR	255	Unique QR code string	Stores the code that the student scans

Table: parent

Field Name	Data Type	Size	Description	Purpose
parent_id	INT	11	Unique number	Primary key to identify each parent
student_id	INT	11	Foreign key	Links the parent to a specific student
contact_number	VARCHAR	15	Phone number	Stores the number where notifications are sent

Table: attendance

Field Name	Data Type	Size	Description	Purpose
attendance_id	INT	11	Unique number	Primary key for each attendance record
student_id	INT	11	Foreign key	Identifies which student attended
time_in	TIME	(none)	Time of scan	Records when the student scanned the Barcode
date	DATE	(none)	Date of class	Records the date of the

				attendance
--	--	--	--	------------

Table 4

3.12 Entity Relationship Diagram

The ERD shows how the tables (entities) in our database are connected to each other.

- **Entities:** STUDENT, PARENT, ATTENDANCE
- **Attributes:** Each entity has fields like student_id, first_name, etc.
- **Relationships:**
 - One **STUDENT** has one **PARENT** (One-to-One relationship). A student is linked to one parent record.
 - One **STUDENT** can have many **ATTENDANCE** records (One-to-Many relationship). A student can log attendance on many different days.

Figure 5

3.13 Prototypes

This section shows simple sketches or sample designs of the key screens in our system. These are not the final designs but help us visualize the layout.

1. Login Page (for Teacher)

- A simple screen with two text boxes: "Username" and "Password".
- A blue "Login" button below.
- A title at the top: "Notification Subsystem - Teacher Login".

2. Teacher Dashboard

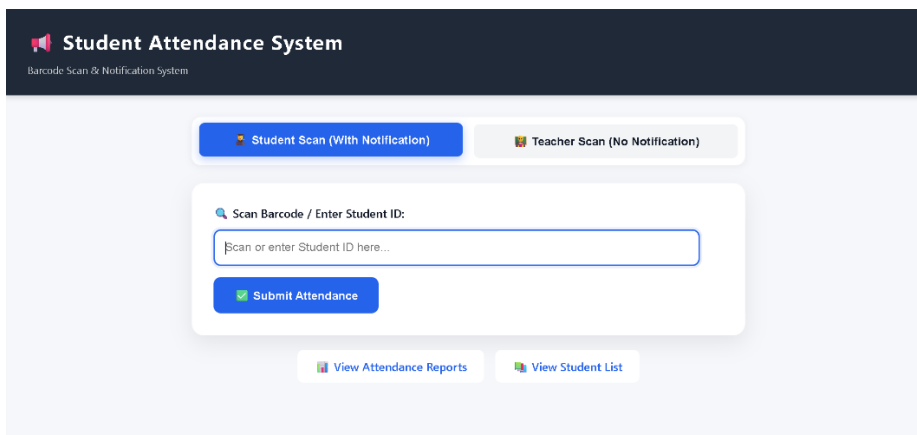
- A header showing the teacher's name and today's date.
- A menu on the left side: "Today's Attendance", "Student List", "Reports".
- A main area showing a list of students with checkboxes for "Present" or "Absent".

3. Barcode Display Screen (for Student)

- A large square in the middle of the screen showing a QR code.
- A label above the code: "Scan this code to log your attendance."
- The student's name displayed below the code.

4. Parent Notification (Sample)

- A mock text message on a phone screen.
- Sender: "Lab A System"
- Message: "Alert: Your child, Juan Dela Cruz, was marked PRESENT in Computer Laboratory A at 8:05 AM."



The screenshot displays the 'Student Attendance System' interface. At the top, a dark blue header contains a speaker icon, the title 'Student Attendance System', and the subtitle 'Barcode Scan & Notification System'. Below the header, there are two buttons: 'Student Scan (With Notification)' in blue and 'Teacher Scan (No Notification)' in light gray. The main area features a white card with a search icon and the text 'Scan Barcode / Enter Student ID:'. Below this is a text input field with the placeholder 'Scan or enter Student ID here...'. A blue button with a green checkmark and the text 'Submit Attendance' is positioned below the input field. At the bottom of the interface, there are two more buttons: 'View Attendance Reports' and 'View Student List', both in light gray.

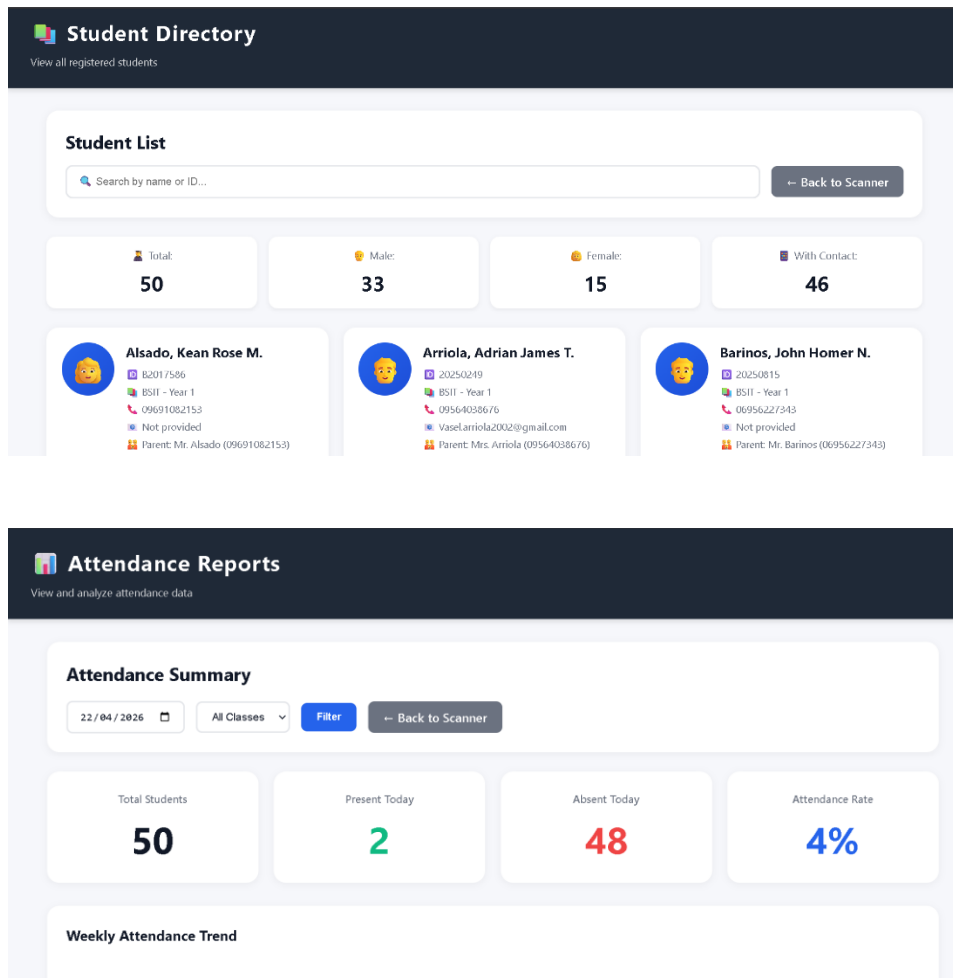


Figure 6

Chapter 4

Results and Discussion

4.1 Results and Discussions

In this part, present the data you gathered from surveys, tests, or system evaluation. Use tables, charts, or figures to show results clearly. Then, discuss what the results mean—relate them to your objectives, and explain any patterns, findings, or unexpected outcomes.

4.2 System Evaluation

In this part, describe how you assessed the performance of your system. Include feedback from users or evaluators using criteria such as usability, functionality, reliability, and efficiency. Present the results of the evaluation (e.g., survey ratings or comments) and explain how well the system met the objectives.

Chapter 5

Summary Findings, Conclusion and Recommendation

5.1 Summary Findings

In this part, highlight the key results of your study.

Summarize the most important findings based on your data and system evaluation. Keep it clear and focused on how your objectives were met or addressed.

5.2 Conclusion

In this part, summarize the overall outcome of your study.

State whether your objectives were achieved, and briefly explain the impact or significance of your system. You may also mention any lessons learned during the development process.

5.2 Recommendation

In this part, suggest ways to improve the system or how it can be implemented effectively.

You may also recommend future enhancements, additional features, or areas for further study based on your findings.

References

List all the sources you used in your research using IEEE Format. Here are sample formats:

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Appendices

Appendix A - Gantt Chart

Appendix B - Communications

Appendix C - Evaluation Tool and Test Document

Appendix D - User Manual

Appendix E - Picture Showcasing the data gathering, investigation, etc.

Appendix F - Proposed System Budget

Appendix G - Curriculum Vitae